

NABL Document Intelligence

Project Overview

Status as of 2026-09-03. Covers both halves of the project: the live 3-phase applicant assistant (backend/ + frontend/) and the offline ground-truth & benchmark pipeline. Every figure was read from the real source tree or a real output file — none is projected or estimated.

Headline: extraction now measures **0.624 key F1 / 54.0% field accuracy** against 53 hand-authored ground-truth documents, using Amazon Nova on Bedrock. Test-table extraction is strong (94–96%); document-level field *recall* is the weak point (36% of fields missed).

Note: supersedes docs/NABL_Full_Project_Report.pdf (2026-09-02), which describes an 8-stage pipeline that has since been removed from the codebase.

1. Overview of this document

NABL Document Intelligence helps a laboratory complete its NABL accreditation or recognition-scheme application by reading the documents it already has, instead of making staff retype everything. It covers all 9 real NABL application forms (151, 152, 153, 153A, 154, 155, 157, 158, 159) and has three phases: an eligibility wizard, a document ingestion pipeline (text/OCR + LLM classification + LLM extraction), and a human-in-the-loop review UI where a person confirms or corrects what was auto-filled.

Alongside the product there is a measurement track: a small offline pipeline that stages raw lab reports for hand labeling, then re-extracts each one with Amazon Nova and scores the result against those hand-written labels. That is where the accuracy numbers in this document come from.

This document reflects a material architecture change made on 2026-09-02/03: the previous 8-stage, LLM-driven dataset pipeline (schema discovery → master schema → labeling → quality control → final dataset → benchmark) was **deleted** and replaced by a 2-script pipeline whose ground truth is written by hand. Section 9 explains why this made the reported accuracy go *down* while making it far more trustworthy.

2. Goal

Let an applicant lab fill most of a NABL application by uploading the certificates, CVs, PT/ILC reports and equipment records it already holds — with a human reviewing anything uncertain before it counts as final — and be able to state, with evidence, how accurate that extraction actually is.

3. Requirements

3a. Functional

- Gate document upload behind a conversational eligibility check: 4 ISO-readiness questions for the 6 full-accreditation forms, a lighter 2-question gate for the 3 recognition-scheme forms. A deterministic state machine decides pass/fail; the LLM only turns a free-text answer into a verdict.
- Accept PDF, DOCX, JPG and PNG uploads, and route each by its real shape: born-digital PDF → PyMuPDF text + true bounding boxes; DOCX → python-docx text; scanned page in devanagari/arabic/ta/te/ka → the deepreef-ocr AWS Lambda; scanned English page → local RapidOCR in-process, with a vision-LLM call only if that fails.
- Classify every upload into one of 17 supporting-document types, or as a `completed_application_form` when the upload is a filled copy of the form itself.
- Extract into the correct Pydantic schema for the chosen form (9 forms across 2 schema bases), with deterministic regex extraction for GST/PAN/TAN ahead of any LLM call.
- For a whole filled-in form: page-level chunking, embedding retrieval per schema section, per-section extraction, then a targeted retry pass for fields that came back empty.
- Extract key/value pairs that have *no* schema slot as well, on every document, and surface them in the review UI beside the schema fields (added 2026-09-02, see Section 7).
- Merge all documents into one compiled form, reporting cross-source conflicts instead of silently picking a winner, and render the result back out as a filled PDF.
- Offline: stage raw documents for hand labeling, re-extract each with Nova, and score predictions against the hand-written ground truth.

3b. Non-functional

- No paid API key required to run the product: the LLM chain tries each configured provider in order and only fails the request if every one fails, returning 503 naming each failure.
- A failed extraction marks that document status: "failed" with a clear error; it never crashes the request or loses other documents' results.
- Every external call (LLM, OCR, AWS, Qdrant) is mocked in the test suite, so the full suite runs with no API key and no live services.
- One unreadable page, one exhausted provider, or one failing schema section must not discard work that already succeeded.
- Reuse existing infrastructure rather than growing a second copy of it — the OCR engine, the LLM chain, and the PDF/text utilities are shared by both the live app and the offline pipeline.
- Report only measured numbers. An unrun stage is reported as unrun.

4. Background

The 9-form schema (`app/schemas/forms.py`, `common.py`) is modeled directly on the official NABL templates, read in full rather than sampled: 151/152/153/153A/154/158 are full ISO/IEC 17025 or ISO 15189 accreditation applications sharing one schema base, while 155/157/159 are lighter PT-performance-based recognition schemes sharing a second, narrower base. NABL 100B is a procedure document with no applicant fields and is not modeled as a form.

The measurement track went through one full rewrite. The first version generated its own "ground truth" with an LLM and filtered it with rule-based checks, which produced a flattering but circular 0.832 F1 — the system was largely being scored against its own earlier output. That entire pipeline was removed. The current version stages each document's real text or page images for a human (or Claude, reading the page directly) to label by hand, and scores against that. All 53 documents in the dataset are now hand-labeled.

5. Who's your customer

- **Primary:** the staff of a laboratory applying for NABL accreditation or recognition — typically a quality manager assembling one of 9 forms plus its supporting evidence.
- **Secondary:** this engineering team, which needs the extraction pipeline's failure modes to be visible (per-field confidence, conflicts, a structured extraction report) rather than hidden behind a single "success".
- **Downstream:** whoever reviews the completed application before submission — the lab's own reviewer today, potentially a NABL assessor receiving a more complete package.

6. Tenets

- Deterministic before LLM wherever the format allows it — the eligibility gate and GST/PAN/TAN extraction are rule-based, not model calls.
- Never crash on a bad document or an unlucky provider outage; degrade to a clear per-document or per-request error.
- Every extracted value should be traceable to a page/region of its source document, not a bare value with no provenance.
- Ground truth is worth doing by hand. A metric measured against generated labels flatters the system that generated them.
- Free-tier friendly: the product must be fully runnable without a paid API key.

7. Design

7a. Challenge

Nine real forms split into two genuinely different shapes, so one flat schema cannot represent them. Supporting documents range from a one-page GST certificate to a 30-page filled form, and from clean born-digital PDFs to photographed pages in five non-English scripts. A whole filled form needs a fundamentally different extraction strategy than a single-purpose certificate — 100K+ characters does not fit reliably in one free-tier call. Free LLM providers are each individually unreliable (rate limits, small context windows, outages), so none can be a hard dependency. And no pre-existing answer key exists for any of it, so accuracy cannot be measured until one is created by hand.

7b. Design options considered

Decision	Options	Chosen
OCR routing	(A) one OCR engine for everything (B) route by script — Lambda for 5 non-English scripts, local RapidOCR for English, vision-LLM as last resort	B
Whole-form extraction	(A) one large call over the full text (B) page-chunk + embed + retrieve top-k per schema section + targeted retry for missing fields	B
Provider strategy	(A) one fixed provider (B) ordered multi-provider fallback chain tried per call (now 11 services incl. Nova on Bedrock)	B
Fields with no schema slot	(A) drop them, as before (B) a separate “extras” panel (C) extract on every document and show them beside the schema fields	C
Ground truth	(A) LLM-generated + rule-filtered (original) (B) hand-authored per document, LLM-free	B

7c. Pros

- **Script-routed OCR:** English (the common case) needs no AWS call at all; non-English scripts still get a purpose-built model rather than a generic engine's guess.
- **Chunk + retrieve + verify:** only the chunks relevant to a section are sent per call, and the retry pass recovers missed fields without re-asking about everything.
- **Multi-provider chain:** works with zero paid spend, and one provider's rate limit or outage does not take the feature down.
- **Open-ended extraction:** a document the 17-type classifier cannot place (previously extracting *nothing*) now still yields usable fields; nothing found in a document is silently discarded for lack of a schema slot.
- **Hand-authored ground truth:** the accuracy number now means something. It is measured against what a human read on the page, not against another model's opinion.

7d. Cons

- **Script-routed OCR:** the vision-LLM last resort has no per-field bounding box, so the review UI can only show a page-level highlight for those fields.
- **Chunk + retrieve + verify:** wired only for the whole-form path, so there are two extraction code paths to maintain rather than one.
- **Multi-provider chain:** free-tier quality, context size and JSON/vision support vary by provider (only Gemini and Nova take images), so which provider answered a given call is not guaranteed.
- **Open-ended extraction:** costs an extra LLM call per ~7,000 characters, has no relevance gate (an unrelated report still contributes fields), and its values land in an untyped `extra_fields` dict that cannot be validated or placed into the official form's sections.
- **Hand-authored ground truth:** does not scale — 53 documents took real human effort, and every new document needs the same before it can be scored.

System map (real modules)

Phase	Module(s)	Role
Wizard	app/wizard/	Deterministic prerequisite gate; LLM only parses free-text answers.
Ingestion	documents/pipeline.py, classifier.py, extractor.py	Route by file type, classify into 17 doc types, extract schema fields + open-ended fields.
Ingestion (whole form)	documents/chunking.py, embeddings.py, vector_store.py, retrieval.py, verification.py	Page-chunk → embed → Qdrant → retrieve top-k → per-section extract → retry missing.
Ingestion (rule-based)	documents/rule_extraction.py	Regex GST/PAN/TAN before any LLM call.
OCR	documents/local_ocr.py, ocr_client.py	Local RapidOCR (English) / deeppreef-ocr Lambda (5 scripts).
Compile	documents/compiler.py, extraction_report.py	Merge into the form shape, route open fields to extra_fields, detect conflicts.
Review (HITL)	routers/review.py, documents/form_pdf.py	Serve compiled form + extraction report, field-patch endpoint, render filled PDF.
LLM chain	app/llm/factory.py, chain.py, providers.py	Ordered fallback across 11 services (14 chain entries incl. key rotation).
Ground truth	scripts/create_ground_truth.py, dataset_normalization/	LLM-free staging of text/page images for hand labeling; text_repair.py fixes glued words.
Scoring	scripts/generate_predictions_and_score.py, app/benchmark/	Nova re-extracts each labeled document; accumulator.py/compare.py score it.
Frontend	frontend/src/	WizardPage, UploadPage, ReviewPage; DocumentViewer, FormField, ChatPanel.

8. Conclusion

Both halves of the project are working and measured. The live app is code-complete across all three phases with 207 tests passing and 0 failing across 22 test files (3,016 lines), every external call mocked. The offline track has all 53 dataset documents hand-labeled, all 53 re-extracted by Nova, and a real scored summary.

- **Extraction never fails outright:** extraction_success_rate 1.00 and domain_accuracy 1.00 across 53/53 documents.
- **Test tables extract well:** test_name 94.4%, test_result 96.2%, unit 97.6%, reference_range 88.3%.
- **Document-level fields are the weak point:** key recall 55.2% (36.2% of ground-truth fields missed), field accuracy 54.0%, and exact_match_rate 0.00 — not one document came out perfectly.
- **Hallucination is comparatively low:** 9.8% of extracted keys do not exist in the ground truth.
- **The headline number went down on purpose:** 0.832 F1 (old, self-scored) → 0.624 F1 (now, against hand-written truth). The system did not get worse; the measurement got honest.

The clear next engineering priority is **recall**, not precision: Nova is reading the results tables well but missing a third of the document-level fields. Section 9 lists the decisions needed before this goes in front of a customer.

9. FAQs

Open challenges to discuss with the lead

Listed as open items only — the answers are what the discussion is for. Ordered roughly by how much each one blocks progress.

#	Challenge
1	Field recall. 36.2% of ground-truth fields are missed, key recall is 0.5515, and exact_match_rate is 0.00 — no document came out perfectly. There is no agreed accuracy bar for a pilot.
2	The reported figure moved from 0.832 to 0.624 F1 when ground truth was rebuilt by hand. Anyone who saw the earlier number has not been told why it changed.
3	Hand-authored ground truth does not scale. The 53 labeled documents took real human effort; no owner or budget is identified for labeling any more.
4	The 8-stage pipeline's removal is unconfirmed. Its 9 scripts, 7 test files and package sources are deleted, but 7 package directories remain as empty shells with stale __pycache__.
5	Almost nothing is committed. 3 commits exist against ~25 modified/untracked paths, including the entire current feature set. No branch or review policy is in place.
6	Open-ended extraction has no relevance gate. An unrelated lab report currently contributes arbitrary fields to a NABL accreditation form.
7	Per-document extraction cost is unbounded. Open-ended extraction costs one LLM call per ~7,000 characters (a 17-page document is ~3-5 extra calls, a 30-page filled form ~10+). No per-document call or cost ceiling exists.
8	extra_fields is flat and untyped. Dynamically discovered fields cannot be validated, typed, or placed into the official NABL form's sections or PDF layout.
9	No per-domain or per-format breakdown in the current scoring summary, so the scanned/image vs born-digital split — where the recall gap most likely concentrates — is invisible.
10	Nova's deployment requirements live only in code comments. It needs a region-prefixed inference-profile ID (bare model IDs are rejected) and an explicit maxTokens (Bedrock's default truncated a reply's JSON mid-object).
11	Docs drift. README states port 8000 and VITE_API_BASE_URL while the frontend uses port 8010 and VITE_API_BASE; README claims 3 LLM providers where 11 are wired.
12	Duplicate fields can reach reviewers. Schema-guided and open-ended extraction can return the same value under different names, and no dedup exists.

10. Appendix

10a. Metrics — scored 2026-09-03, Nova on Bedrock vs. hand-authored ground truth

Metric	Value	Reading
Documents scored	53	Every document in the dataset; all hand-labeled.
extraction_success_rate	1.0000	No document failed to produce a prediction.
domain_accuracy	1.0000	Document domain classified correctly every time.
key_precision	0.7181	Of the keys extracted, ~72% exist in ground truth.
key_recall	0.5515	Of the ground-truth keys, only ~55% were found.
key_f1	0.6239	Headline figure.
field_accuracy	0.5396	Of keys both sides have, ~54% of values match.
exact_match_rate	0.0000	No document was extracted perfectly end to end.
missing_field_rate	0.3617	Dominant failure mode.
hallucination_rate	0.0979	Comparatively low — it under-reports rather than invents.
value_accuracy	0.7054	
unit_accuracy	0.9756	Units are read reliably.
reference_range_accuracy	0.8826	
test_name_accuracy	0.9436	Results tables are the system's strength.
test_result_accuracy	0.9623	
test_unit_mapping_accuracy	0.9756	

Source: *predictions/summary.json* on the shared drive. No per-domain or per-format breakdown is currently produced (see Section 9, item 9).

Codebase

Area	Count
Tests	207 passed, 0 failed — 22 test files, 3,016 lines
documents/ (ingestion, OCR, RAG, compile)	20 files, 2,060 lines
llm/ (11 services, 14 chain entries)	6 files, 746 lines
schemas/ (9 NABL forms, 2 bases)	4 files, 627 lines
dataset_normalization/ (staging + text_repair)	7 files, 585 lines
benchmark/ (accumulator, compare, models)	5 files, 541 lines
routers/ (12 REST endpoints)	5 files, 455 lines
wizard/	3 files, 186 lines
frontend/src	9 files, 928 lines
Entry-point scripts	3 — create_ground_truth.py, generate_predictions_and_score.py, upload_document.py

Area	Count
Empty package shells awaiting cleanup	7 — kv_extraction, labeling, master_schema, quality_control, schema_discovery, final_dataset, final_dataset_qc
Git	3 commits; ~25 modified/untracked paths uncommitted

10b. Performance

Aspect	Note
Per-request LLM/OCR timing	Not centrally logged. Per-provider timeout is configurable (llm_timeout_seconds, default 30s).
Nova (Bedrock)	maxTokens must be set explicitly (default 8192 here); without it Bedrock truncated a dense document's JSON mid-object. Multimodal, so scanned pages go as images rather than OCR text.
Ollama (local)	210s timeout — far longer than any hosted provider's 30s, consistent with local CPU inference being the slowest path.
Retrieval	all-MiniLM-L6-v2 embeddings (~90MB, 384-dim), top_k=4 chunks per schema section, on-disk Qdrant with no server process.
Open-ended extraction	One call per ~7,000 characters (OPEN_EXTRACTION_CHUNK_CHARS), capped at 40 fields per chunk.
Confidence threshold	0.85 default — the line the review UI uses to flag a field for manual attention.

10c. Edge cases handled

- **Every configured provider fails on one call** — 503 naming each provider's specific failure, not a silent 500.
- **A mostly-digital PDF with a few scanned pages** — checked and OCR'd per page, not per document.
- **A long multi-page document** — open-ended extraction chunks it, after a real bug where a single truncated call silently ignored everything past ~page 4 of a 17-page file.
- **An upload matching no known doc type** — classified other, and (since 2026-09-02) still extracted rather than returning nothing at all.
- **A whole filled form exceeding a provider's context** — fails with a clear “every provider failed” error instead of a silently truncated extraction.
- **Two documents disagreeing on one field** — recorded as a conflict in the extraction report, not silently overwritten.
- **Glued-together OCR/PDF text** (“Thisistocertify...”) — repaired by dataset_normalization/text_repair.py.
- **Concurrent uploads hitting the local vector store** — one shared, lock-guarded Qdrant client, since local-mode Qdrant holds a file lock.

10d. Glossary

Term	Meaning
doc_type	One of 17 categories an uploaded supporting document is classified into, or completed_application_form when the upload is a filled copy of the form itself.
FormCategory	ACCREDITATION (151/152/153/153A/154/158, full ISO application) or RECOGNITION (155/157/159, lighter PT-performance scheme).

Term	Meaning
Ground truth	What a human read on the page, written by hand into labelled_dataset//.json. No LLM involved in producing it.
Open-ended extraction	An extra LLM pass that returns key/value pairs under field names it invents, with no schema whitelist — runs on every document.
extra_fields	The flat dict on the compiled form where open-ended fields land, since they have no typed schema slot.
Grounding	Tying an extracted value back to the page/region of the source document it came from.
Key precision / recall / F1	Of the keys extracted, how many are real (precision); of the real keys, how many were found (recall); their harmonic mean (F1).
Field accuracy	Of the keys both prediction and ground truth contain, the fraction whose values also match.
Hallucination rate	The fraction of extracted keys that do not appear in ground truth at all.
HITL	Human-in-the-loop — the review step where a person confirms or corrects auto-filled fields.
Nova	Amazon Nova on AWS Bedrock, via the Converse API; authenticated by the ambient AWS credential chain, not an API key.